

Team 1 Research

how are hardware and software used within the context of the industry?

Within the IT and digital solutions industry, hardware and software interact through a client-server cloud architecture to deliver scalable, accessible platforms. The backend relies on cloud data centres and server clusters (hardware) managed by virtualisation software. This combination allows the system to auto-scale, automatically provisioning more processing power and memory when student or school traffic spikes, ensuring the platform remains stable without requiring manual hardware intervention.

We will use different hardware and software to make our website work. The website will be hosted on servers which will provide the processing power and storage needed. Users will be able to access the website on laptops, tablets and phones. We will use HTML and CSS for the front end, and we could use JavaScript for some interactive features. We could use PHP for the back end and MySQL to store things like user accounts and expressions of interest.

Summary

We will use servers, HTML, CSS, JavaScript, PHP and MySQL to build and run our website.

We also need to make sure our software works with different hardware and accessibility devices. For example, some users might use screen readers if they have sight loss. We will make sure our website is designed so these tools can understand the information on the page. We will also make sure the website works on different screen sizes so people can use it on the device they have.

Summary:

We will make our website work with different devices and accessibility tools.

colouring of the solution:

Also add white with black text so the contrast is higher for poor sighted people -

AI is used in the industry in various ways such as using it to create code or to get help and advice with certain scenarios. However, doing this may save time and help with work but it would also make the user too reliant on the AI which wouldn't let the user learn or understand the code so they might not see the mistakes because AI is not perfect it also makes mistakes. Also, they are used to make complete websites and just

to create literally anything like a new google but can make everything for you basically spoon feeding the answers to you.

We could also use AI in our website by having a chatbot. The chatbot could answer questions about T Levels and Amazon placements. This could make it easier for users to find information instead of having to search through lots of pages. It could also be available at any time, which means users would not have to wait for someone to answer their question.

Summary

We could use an AI chatbot to help users quickly find information about T Levels and Amazon placements.

There are some problems with using AI as well. AI can sometimes give wrong information, so we would need to check the answers it gives. We could also look at other technologies such as touchscreens, virtual reality and text-to-speech. We would need to decide which technologies would actually be useful for our users instead of just adding them because they are new.

Summary

AI can be useful but can make mistakes, so we need to check it and only use technologies that benefit our users.

The Users Needs

We need to think about who will use our website. Students will need information about T Levels, Amazon placements and resources that can help them. Schools and teachers might need information that they can give to students. Parents and carers could also use the website to find out more about T Levels and the opportunities available at Amazon.

Summary

We will design the website around students, schools, teachers, parents and carers.

We also need to think about people who have disabilities. We could add things like larger text, good colour contrast, keyboard controls and screen-reader support. We could also add alt text to images. This would make our website easier for more people to use and would help us meet the accessibility requirements in the brief.

Summary

We will add accessibility features so people with different disabilities can use our website

The industry-specific guidelines and regulations you will need to follow

We will need to follow data protection requirements because our solution could collect personal information from users. We will only collect information that we actually need and we will make sure it is stored securely. We will also make sure users can keep their information up to date and, where appropriate, request for their information to be removed. We will restrict access to personal information so unauthorised people cannot access it.

Summary

We will only collect necessary personal information and will keep it secure.

We will also consider WCAG accessibility guidelines when designing our website. We will use alt text for important images so screen-reader users can understand them. We will also use suitable colour contrast and make sure our website can be used with a keyboard. We will make sure our content is clear and readable and that our website works across different screen sizes.

Summary

We will follow accessibility guidelines so our website can be used by as many people as possible.

Best languages

HTML and CSS for frontend as they are the main options for web design.

Backend – JavaScript can be run client side; therefore, it requires less server-side processing power. However, JavaScript gives all the code to the user's device, therefore meaning all the code is easily accessible and modifiable by the user. This makes it harder to produce secure code in this language, and easier for malicious actors to find vulnerabilities due to it being all available for them to see. PHP would be better as it keeps all the code server side, therefore harder for people to find vulnerabilities. Overall PHP is better option.

A website is better than an app as the team members is more familiar with HTML CSS and PHP than any languages like C#.

Sign up system

Using PHP to interact with the database, use MySQL for the database that to store user credentials.

Will need a separate page for sign up and login.

Password does not need to be encrypted before the request is sent on sign up as https can handle all the encryption, however once it arrives on the web server it should be hashed before being stored in the database.

When login requests are sent it should be hashed before sent over the web, then the hashes can be compared.

No need to overcomplicate it, can use the same sign up and log in page, with 2 buttons for sign up and login.

Add a password reset option if have time, given using emails or text messages would be too much for this project though, it would likely either must have a security question, or no security.

Social features

Add links that can be shared too specific parts of the webpage.

For example, have one div that includes an image title and a paragraph, and a share button at the bottom to automatically move someone to that part of the page if they use that link.

Content that you must sign up for

Use PHP to make one webpage only accessible to people who are logged in, this could have slightly more specific information on there.

Downloadable content

Have a pdf version of the webpage be downloadable so that they can access it offline.

Have a pdf with t level info be downloadable.

colouring of the solution: -

Amazon Orange (#FF9900) and Amazon Dark Blue (#232F3E) are the primary brand colours - billy

Types of accessibility features examples:

- . text to speech
- . speech to text
- . easy inputs, drop down boxes etc
- . different sized fonts for easy viewing
- . correct colour contrasting for easy viewing of text

the industry-specific guidelines and regulations you will need to follow:

WCAG

Pep-8

IT compliance standard

CTA'S

Current technologies within the industry:

Touch screen

Virtual reality

Artificial intelligence

Accessibility features - text to speech, colourblind modes et

How are these technologies used within industry:

Accessibility features are used for those who have disabilities and cannot access all of the normal features of the system meaning they will not be able to use certain websites or systems. Accessibility features make it easier use for those with disabilities meaning they will be able to utilise those systems just like anyone else.

How has AI technology affected the digital industry?

AI has affected the digital industry by reducing the need for entry level jobs as it can create code at the level of an entry level worker, and debatably better, however it also creates more of a demand for higher level jobs for the people who then go reviewing the vibe-coded code, or also the people who create, maintain and manage the AI models, making the tech industry harder to get into, but also increasing demand and skillsets required for the job.

For the social feature aspects I think we could add some form of image/text posting feature, where users can share linked in style posts with each other

How digital solutions could be used to meet different user needs – Accessibility should be built in mind such as a larger surface area to click on for users with motor needs. For users with eyesight needs, the background colour scheme should be dark, so it does not contrast too much and strain their eyes trying to look while doing their T Level placement. Also allow a magnifying mouse for any text that would need zooming in. There should also be other types of websites provided by Amazon to help people on T Level placements such as where they can code such as GitHub or JavaScript for any programming languages. As well as more information on the main site such as the schedules and briefs for their emails.

how digital solutions could be used to meet different user needs

You can use the users for feedback by using the surveys and reviews. This is to try and find out how the user acts and how the user will use the website there are many other things they will be looking for. We also need to figure out the target audience of the website so we can design it a certain way.

Newly emerging technologies – An example of a much more newly emerging technology would be using Zoom for online meetings that have become more popularized and common place than ever since the Covid-19 lockdown which makes it easier in saving time from having to go to an Amazon employee centre so they can host the meeting from their own homes and feel more comfortable while doing so. There are websites and emails that will save their data. AI can also be used to help plan out certain events or tasks for them to focus on working on to increase their employability skills in their experience through chat boxes to ask for any help or just communicate with other t level students on the placement or even their employers to mentor for them in case.